UNIT – V: Multithreading and Event Handling

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Main Thread-Creation of New Threads, Thread PrioritySynchronization, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads. Event Handling- Events, event sources, event classes, event listeners, delegation event model, handling mouse and keyboard events.

```
import java.awt.*;
import java.applet.Applet;
import java.awt.event.*;
/*<applet code="ButtonDemo.class" height=500 width=500>
</applet>*/
public class ButtonDemo extends Applet implements ActionListener
{
	Button b1,b2,b3;
	public void init()
	{
		b1=new Button("Orange");
		add(b1);
		b2=new Button("Green");
		add(b2);
		b3=new Button("Black");
		add(b3);
		b1.addActionListener(this);
		b2.addActionListener(this);
		b3.addActionListener(this);
	}
	public void actionPerformed(ActionEvent e)
	{
		String msg=e.getActionCommand();
		if(msg.equals("Orange"))
			setBackground(Color.ORANGE);
		else if(msg.equals("Green"))
			setBackground(Color.GREEN);
		else if(msg.equals("Black"))
			setBackground(Color.BLACK);
		repaint();
	}
}
```

//java program to handle key events

```
import java.awt.*;
import java.awt.event.*;
import java.applet.Applet;
/*<applet code="KeyDemo.class" width=500 height=500>
</applet>*/
public class KeyDemo extends Applet implements KeyListener
{
	String msg="";
	public void init()
	{
		addKeyListener(this);//registration
	}
	public void keyPressed(KeyEvent e)//implementation method1
	{
		showStatus("U are pressing the key");
		repaint();
	}
	public void keyReleased(KeyEvent e)//implementation method2
	{
		showStatus("U released the key");
		repaint();
	}
	public void keyTyped(KeyEvent e)//implementation method3
	{
		msg=msg+e.getKeyChar();
		showStatus("U typed some text");
		repaint();
	}
	public void paint(Graphics g)
	{
		g.drawString(msg,250,250);
	}
}
```

//java program to handle mouse events

```
import java.awt.*;
import java.applet.Applet;
import java.awt.event.*;
/*<applet code="MouseDemo.class" height=500 width=500>
</applet>*/
public class MouseDemo extends Applet implements
MouseListener,MouseMotionListener
{
	public void init()//registration process
	{
		addMouseListener(this);
		addMouseMotionListener(this);
	}
	public void mouseClicked(MouseEvent e)//The following 5 are mouse listener
methods
	{

		showStatus("Mouse Clicked at "+e.getPoint());
		repaint();
	}
	public void mousePressed(MouseEvent e)
	{

		showStatus("Mouse Pressed at "+e.getPoint());
		repaint();
	}
	public void mouseReleased(MouseEvent e)
	{

		showStatus("Mouse Released at "+e.getPoint());
		repaint();
	}
	public void mouseEntered(MouseEvent e)
	{

		showStatus("Mouse Entered at "+e.getPoint());
		repaint();
	}
	public void mouseExited(MouseEvent e)
	{
```

```
		showStatus("Mouse Exited at "+e.getPoint());
		repaint();
	}
	public void mouseMoved(MouseEvent e)//The following 2 methods are mouse
motion listener methods
	{

		showStatus("Mouse Moved at "+e.getPoint());
		repaint();
	}
	public void mouseDragged(MouseEvent e)
	{
		showStatus("Mouse Dragged at "+e.getPoint());
		repaint();
	}
}
```